

06

LECTURE

HEALTH CARE

MT 1, Second Semester LECTURE 06 - Sir Micheal Gu, RMT



ENVIRONMENTAL HEALTH PROTECTION: RESIDENTIAL, OCCUPATIONAL, AND RECREATIONAL ENVIRONMENT

TOPIC OVERVIEW

I. Residential Environment

- A. Housing and health
- B. Basic principles of healthful housing
- C. Healthy housing
- D. Substandard housing
- E. Slum
- F. Problems associated with poor housing
- G. Aspects of housing quality: Disease vectors
- I. Housing standards— Philippines

II. Occupational Environment

- A. Right to work
- B. Definition of terms
- C. Brief history
- D. International Labour Organization
- E. Occupational health and safety in the Philippines
- F. Hazard
- G. Occupational Hazards
- H. The OSH Management Cycle
- I. Occupational Health Surveillance
- J. Role of Occupational Health Services
- K. Multidisciplinary Teams
- L. Hierarchy of Controls
- M. Health Promotion, Education, and Training
- N. Total Worker Health

III. Recreational Environment

- A. Recreational areas
- B. Negative impact of recreation
- C. Recreational water environment
- D. Management of recreational areas
- E. Recreation in the age of Covid-19
- F. Management and Covid-19 considerations

LEARNING OBJECTIVES

1. Correlate the health of community members with the state of their dwellings.
2. Discuss the impact of the workplace on worker health.
3. Enumerate methods to enhance safety in recreational areas.
4. Identify key OSH principles and legal frameworks.

I. RESIDENTIAL ENVIRONMENT

- "A house is not a home unless it contains food and fire for the mind as well as the body." -*Benjamin Franklin*
- A house **must be more than just a physical shelter**—in terms of environmental health, we view our residences as our primary sanctuaries. Individuals should be able to leave the pressures of society behind and relax in safety.

A. HOUSING AND HEALTH

- Where we live, the house or the neighborhood, has an impact on our health.
- **Residence** refers to one's house or living accommodation
 - Can be temporary (e.g., Dormitories) or permanent accommodation
- **Quality of Housing** refers to the physical condition of an individual's house, as well as the quality of the social and physical environment in which the house is located

- Ex. Good neighbors, Status of the area in terms of pollution

- **Healthy Housing** refers to a shelter that supports a state of complete physical, mental, and social well-being (WHO).
 - A non-negotiable element in a healthy community.
 - It means having proper heat, light, ventilation, and plumbing, all of which can directly affect a person's emotional and physical state.

B. BASIC PRINCIPLES OF HEALTHFUL HOUSING

- We address the fundamental needs of a human when it comes to housing.
- **4 Fundamental Pillars**
 - Meet physiologic needs
 - Address the human body's internal department
 - Meet psychological needs
 - Mental needs of a person
 - Protection against Diseases
 - Protection against Injury

C. HEALTHY HOUSING

Housing should provide for the following needs:

Physiologic Needs

- **Protection from the elements** (rain, wind, etc.)

- **Protection from undue heat loss**
- **Permit adequate heat loss**
 - Houses should have a stable thermal environment
 - **Housing Guidelines:**
 - Centers for Disease Control (CDC) Housing Code: Indoor temperature should be at least **68°F (20°C)** measured 18 inches above floor level. We have a comfortable temperature for us to live in. Special Cases: **70°F** may be required for those with **specific physical conditions** or advanced age.
- **Proper Ventilation**
 - The window area should **comprise at least 8% of a room's floor space**
 - Windows must open to allow natural air exchange. (eg. opened windows can compensate for a broken a/c)
- **Proper natural and artificial illumination**
 - The needs (proper ventilation and proper natural and artificial illumination) are our light and air needs.
 - Vital to physical health and visual clarity
- **Protection from excessive noise**
 - Noise levels inside the home should not exceed **70 decibels** to prevent long-term hearing impairment.
- **Adequate space**
 - Prevent physiological stressors of overcrowding
- **Sanitary Water Service**
 - Plumbing must be designed with minimal bends to maintain water pressure (engineering side). Slow

pressure directly affects the hygiene of the people in the household

- The lines of water should be buried at least four feet deep to prevent freezing (does not apply to the Philippines)

Psychological Needs

- **A home must be a psychological refuge:** providing adequate privacy, having social family activities, and social community activities.
 - There is a direct correlation between house size and privacy. **As family size decreases relative to the house size, individual privacy increases**
 - **Guidelines:** One room for each individual (except couples/small children), can be a space to contemplate, do homework, etc.
- Psychological needs of healthy housing include **performance of household tasks, the maintenance of cleanliness, aesthetic satisfaction, and concordance with social standards.**
 - The layout of the house should make it easy for us to do our chores (e.g., throwing trash, sweeping the floor, etc.).
 - Linked directly to the aesthetics of the house. We want our houses to be aesthetically pleasing. It is not a luxury but a necessity in maintaining our mental health.
- **Our house should support our mental health**
 - **Guidelines for Mental Health:**
 - Bedrooms/bathrooms accessible through halls,

not other bedrooms. This creates social friction and violates privacy

- **Accessible community facilities.**
 - People living in dormitories often deal with privacy violations due to shared or common bathrooms. Additionally, it can cause stress, especially during early mornings when a lot of people are lining up for the bathroom.
- TL;DR: A household must have good flow for better mental health.

Protection against diseases

- Disease **prevention is a hallmark** of environmental health.

Needs:

- **Sanitary water supply.**
- **Protection against pollution.**
 - a plumbing system that can prevent **sewage from mixing with drinking water**
- **Provide Toilet facilities**
- **Sewage contamination protection.**
 - We can **eliminate the threat** of many waterborne pathogens.
- **Unsanitary conditions near the dwelling**
- **Exclude vermin**
 - No stagnant garbage in front of the house to **prevent the attraction** of vermin, which carry diseases.
- **Facilities for food storage**
 - Have proper facilities in our home to **reduce foodborne illnesses.** (refrigeration)
- **Sufficient sleeping rooms**
 - **Enterobius vermicularis** Pinworms/threadworms (**Kigwa**) are easily transmissible in households where people are sleeping together in the same bed. KIGWA ning magtuon.

Protection against injury

- Injury prevention in a household involves simple physical modifications.

Guidelines:

- Implementation of Building Code
- Handrails on all stairs.
- Locks/bars on upper ground windows.
- Non-skid strips/grab bars in bathrooms.
 - Building codes often **mandate handrails on stairs and non-slip surfaces in the bathroom.**
 - Considers senior citizens in homes by installing non-slip tiles or rubber mats to **protect the elderly against injury.**
 - **Childproofing the home** by covering sharp corners with soft materials (e.g., foam, rubber, or cloth). **Buhaton ni aron di mabukol ang bata.**

D. SUBSTANDARD HOUSING

Definition

- Does not meet the standards of living
 - If the needs (mentioned earlier) are not met, then the housing is substandard
 - Poses an active risk to the physical well-being of the occupants
 - A house that fails to provide basic necessities. (eg. plumbing, and electricity)

Characteristics

- Lack of proper heating, plumbing, electricity, or sanitation.
- Violates the building code

Deterioration

- Deterioration: Housing that is damaged, unmaintained, or structurally unsound.

- Deteriorated housing condition
 - Ex. If there are appliances that no longer work or need replacing, or damage in the house due to old age or overuse.

Health Risks of Substandard Housing

- **Physical Well-being:** Higher risk of infectious disease and respiratory issues.
- **Mental Health:** Decreased mental health due to chronic environmental stress.
- **Community Impact:** Poses a risk not just to occupants, but to neighbors and visitors as well.
- Living in substandard homes will create a ripple effect of risk. Lack of hygiene leads to diseases, and the constant stress of living in an unsafe structure will degrade the mental health of your household
- In the most extreme form, substandard housing is usually found in the slums.

E. SLUM

- A heavily populated urban informal settlement.
- According to international standards, a slum lacks at least one of the four essentials of a healthy house:
 - Access to safe water.
 - Access to sanitation.
 - Safe and secure tenure.
 - Durable housing.
- **A Slum area is characterized by extreme squalor and overcrowding.**
 - Poor housing creates a cascade of risks. Overcrowding makes it impossible to maintain hygiene. Poor hygiene leads to diseases. Additionally, living in an overcrowded environment will

create massive psychological stress.

F. PROBLEMS ASSOCIATED WITH POOR HOUSING

- **Inadequate Lighting**
 - Leads to accidents and hinders cleanliness; few dwellings maximize natural light.
 - Light is essential for hygiene, making us aware of the mess we have in our surroundings.
- **Higher Stress**
 - Poverty and poor housing reduce the body's ability to fight infection.
 - Residents will be more susceptible to pathogens.
- **Chronic Illness**
 - Environmental stress accelerates the deterioration of existing conditions.
- **Food and Water Contamination**
 - Poor hygiene leads to household contamination. Pathogens flourish when there are no proper facilities for washing hands or cleaning food.
 - **Common pathogens:** Salmonella, Staphylococcus, Streptococcus.
 - **Impact:** Higher infant mortality, poorer nutritional health.
- **Respiratory and Eyesight Problems**
 - **Poor Indoor Air Quality:** Contaminants are often higher indoors than outdoors due to a lack of dispersion.
 - Usually, in slums, residents prioritize roofs over ventilation. This leads to improper ventilation and humid environments, causing mold to grow.

- **Inadequate Lighting:** Leads to eye strain and increases the risk of accidents.
 - This also means you cannot see the hazards causing injury.

G. ASPECTS OF HOUSING QUALITY: DISEASE VECTORS

Disease Vectors

- These are beings that carry diseases.
- Some standard housing has cracks in the walls and other conditions that allow pests to enter and colonize our living spaces.

Vector	Disease
Fleas	• Bubonic Plague • Murine Typhus
Cockroaches -particularly dangerous in urban housing because of their droppings and shells, which trigger asthma attacks	Associated with: • Salmonella, • Entamoeba • Polio • Major Asthma Trigger.
Rodents	• Leptospirosis • Rat-bite Fever

Table 1. Disease Vectors

H. HOUSING STANDARDS – PHILIPPINES

Republic Act No. 6541

The National Building Code of the Philippines.

- **Scope**
 - Design, location, siting, construction, alteration, repair,

conversion, use, occupancy, maintenance, moving, and demolition

- **Objective**
 - To ensure public safety and quality of life
- **Maintenance**
 - Regular maintenance checks are a must for all buildings
 - Maintenance to avoid the deterioration to substandard quality
- **Dangerous sites**
 - Property should be at a safe distance from the following elements:
 - polluted streams or bodies of water
 - volcanic sites
 - any type of land that can be a source of an explosion
- **Dangerous buildings**
 - The code defined the characteristics of a dangerous building. A house considered dangerous should be vacated right away
- **Coordination with the Local Government Units (LGU)**
 - The code enforces cooperation between the owners and the LGU.

II. OCCUPATIONAL ENVIRONMENT

- "The real enemy of safety is not non-compliance but non-thinking." – *Rob Long*
- True occupational safety requires critical thinking and the active anticipation of risks.

A. DEFINITION OF TERMS

- **Occupational Health:** Focuses on well-being and illness prevention; professionals are "Industrial Hygienists".
- **Occupational Safety:** Focuses on injury prevention; professionals are "Safety Engineers".
- **OSH:** Science of anticipation, recognition, evaluation, and control of workplace hazards.
- Health and safety are closely related but distinct concepts. **Health** focuses on long-term well-being and the prevention of illnesses, while **safety** is concerned with preventing immediate accidents and injuries. By addressing both health and safety together, we aim to protect not only workers but also the surrounding community and the environment.

B. THE AIM OF OCCUPATIONAL HEALTH

- **International Labor Organization (ILO/WHO)**
Joint Committee Definition: "The promotion and maintenance of the highest degree of physical, mental, and social well-being of workers in all occupations".
- **Requirements:**
 - Safe and healthy work environments.
 - Access to occupational health services for all.

C. RIGHT TO WORK

- **Universal Declaration of Human Rights (Article 23):**
 - Every human has the right to work.
 - Right to a fair wage.
 - Right to join a trade union.

D. BRIEF HISTORY

- **4 B.C.**
 - **Hippocrates** described lead poisoning in miners
- **1556**
 - **Agricola** noted diseases and accidents among miners
- **1567**
 - **Paracelsus** describes smelter worker diseases.
- **1700s**
 - Ramazzini examines health conditions of 50+ occupations.
 - father of occupational medicine
- **1802**
 - **Health and Morals of Apprentices Act**
 - The Industrial Revolution was a time when we saw an increased number of people coming to the cities because there was a higher number of jobs being offered. This led to overcrowding and diseases due to poor hygiene practices. This problem led to the establishment of the Act.
- **1919**
 - **Creation of the International Labour Organization (ILO)**
 - Philosophy: Prevention and protection through international standards
 - Global necessity
- **1970 (OSH in the United States)**
 - The Occupational Safety and Health Act (OSH Act) was a turning point. Created two distinct bodies:

- Occupational Safety and Health Association (OSHA)
- National Institute for Occupational Safety and Health (NIOSH)
 - OSHA: Enforces standards through training and education.
 - NIOSH: A research body that develops and recommends standards.
- **NIOSH and Research**
 - NIOSH: U.S. federal agency for OSH research.
 - Responsible for related research, education, and training programs.
 - Data used to develop safety standards and hazard evaluations. It is their data that helps set the threshold limit values or the exposure limits. (eg. limit of chemical exposure in the workplace.)
- **NIOSH vs. OSHA**
 - NIOSH (The Watchdog): Focuses on research and chronic health issues; identifies chemicals or diseases harming workers.
 - OSHA (The Regulator): Enforces safety and adopts new standards based on NIOSH data.
- **Collaboration: NIOSH identifies the need for change; OSHA implements it.**

E. OCCUPATIONAL HEALTH AND SAFETY IN THE PHILIPPINES

Philippine OSH: Republic Act No. 11058

Occupational Safety and Health Standards Act

- Puts the primary responsibility on the employer. **Employers** are legally required to **provide appropriate safety equipment**, such as helmets and masks. They must also ensure that all workers are properly trained on the hazards they may encounter. For any machine or equipment used, **workers should receive adequate training** beforehand.

Employee Rights and Duties:

- Right to know inherent risks
- Right to refuse unsafe work
- Duty to report accidents
- Active participation in safety planning

F. HAZARD vs INJURY

- **Hazard:** Something with the potential to cause harm (potential hazard)
- **Occupational Hazard:** Risk experienced specifically in the workplace.
- **Goal:** Identifying hazards before they cause harm allows for prevention

G. OCCUPATIONAL HAZARDS

- **Biological:** Influenza, bites, blood-borne diseases
- **Chemical:** Silica dust, engine exhaust, welding fumes, carcinogens
- **Psychosocial:** Burnout, stress, violence, harassment

Type	Description	Examples
Physical Hazards	Environmental factors that can cause harm with or without contact	Radiation, extreme temperatures, high noise levels, and pressure extremes
Ergonomic Hazards	Strains and stresses on the body caused by workplace motions and body positions	- Poorly designed desks, repeated motions (typing), heavy lifting - Repeated trauma is the leading cause of nonfatal workplace illnesses
Safety Hazards	Conditions causing immediate injuries or death	- Slip, trip, and fall hazards, electrical hazards, and unguarded machinery - Falls and roadway injuries are the top causes of workplace fatalities
Psychosocial Hazards	Caused by interacting with people and social conditions at work	Occupational burnout, depression, and workplace violence
Chemical Hazards and Biological Hazards	Dangerous hazards that can cause long-term health effects.	- Chemical: Silica dust, engine exhaust, welding fumes, and carcinogens - Biological: Influenza, blood-borne pathogens (HIV/Hepatitis), or animal bites

Table 2. Types of Occupational Hazards

H. THE OSH MANAGEMENT CYCLE

- **Policy:** OSH policy and participation
 - Involve the workers when making the policy.
- **Organizing:** Accountability and training
- **Planning:** Initial review and implementation
- **Evaluation:** Audit and management review
- **Improvement:** Continual corrective action

I. OCCUPATIONAL HEALTH SURVEILLANCE

- **Purpose:** Early identification of dangers before they cause incurable disease.
 - It is to make sure we have an **Early Warning System (EWS)**.
 - Conducting **regular medical exams** for the workers to catch problems when they are still **beatable**.
- **Methods:**
 - **Environmental assessment:** Identifying factors in the workspace
 - **Biological monitoring:** Testing blood/urine for chemical exposure
 - **Medical surveillance:** Ongoing health exams for workers

J. ROLE OF OCCUPATIONAL HEALTH SERVICES

- **Advisory Role:** Advise on workplace design and choice of machinery
- **Monitoring:** Monitor worker health in relation to their specific tasks
- **Response:** Organize first aid and emergency treatment
- **Adaptation:** Try to ensure the work is adapted to the worker

- **These are the people who are doing the health surveillance:**

- **Internal Consultants:** They're not just treating the injuries, but they also participate in the design of the floor of the factory to ensure that every machine and every task is as safe as possible.

K. MULTIDISCIPLINARY TEAMS

- **In the case of laboratories, the following are included:**
 - Medical Technologists
 - Pathologist
 - Director
- **Composition:** Occupational physicians, hygienists, ergonomists, and OSH nurses (**in the case of larger organizations**)
 - Staff must enjoy professional independence from employers and workers
 - It takes **18 people**, all of whom must be free to make unbiased recommendations. They have the freedom of making as harsh a comment as possible without getting fired, as long as those comments are directed toward occupational, safety, or health.

L. HIERARCHY OF CONTROLS

- **Elimination:** Removing the hazard
- **Substitution:** Using safer materials
- **Engineering Controls:** Redesigning the area
- **Administrative Controls:** Changing work habits
- **PPE:** Last line of defense
- Example: The sharp blade of a microtome cannot be eliminated or replaced because they are essential for the job. Instead, safety

is managed through the hierarchy of controls: engineering controls such as blade guards, administrative controls such as SOPs and restricted access, and proper training. PPE alone is not enough to prevent injury in this case, so multiple safety measures must be used together to reduce risk.

M. HEALTH PROMOTION, EDUCATION, AND TRAINING

- **Purpose:** Organizational investment for the future
- Training must be supported by practical materials and understood by workers
- Lifestyle factors (diet, smoking) should be integrated

N. TOTAL WORKER HEALTH

- **Definition:** Integrating protection from work hazards with promotion of injury/illness prevention.
- **Holistic Approach:** Recognizing that workplace factors contribute to health problems previously considered unrelated to work.
- **Goal:** To advance worker well-being overall
 - Their safety does not just save lives, but also improves the economy and the community as a whole.

III. RECREATIONAL ENVIRONMENT

- "People who cannot find time for recreation are obliged sooner or later to find time for illness."– *John Wanamaker*
- **Recreation** is essential for stress moderation and weight control. However, if these inventions aren't fully managed, they become leading grounds for injury and diseases.

- **Recreation**
 - activities chosen during leisure time to refresh the mind and body.
- **Leisure Time**
 - Time when one is not working.
- **Recreation Area**
 - Land and water areas dedicated to public enjoyment
 - It is a vital part of public health because it involves providing an emotional release from the day-to-day stresses of home and work. It helps us maintain our mental health, but only if the area itself is **sanitary and safe**.
- **Types of Recreational Areas**
 - Parks and campgrounds
 - Picnic areas and resorts
 - Hotels and cabin camps
 - Swimming, hunting, and touring facilities
 - Each type of recreational area can provide environmental health challenges. (e.g., waste disposals, pool sanitation)
- **Recreational Activities and Facilities**
 - Facilities for activities like:
 - Swimming
 - Fishing
 - Hunting
 - Boating
 - Sailing
 - Hiking
 - Picnicking
 - Camping
 - Sightseeing

A. RECREATIONAL AREAS

- **Indoor**
 - **Enclosed Spaces:**
 - Accommodate a limited number of people.
 - They allow for control of air quality and temperature.
 - **Management:**
 - Easier to control and maintain
 - Examples: Gymnasiums, Indoor pools, cinemas
- **Outdoor**
 - **Open Spaces:**
 - Harder to control due to size and natural variables.
 - It is safer in terms of air quality due to exposure to fresh air.
 - **Venue Risks:**
 - Recreation and sports venues are actually more common sites for injury than schools or workplaces.
 - Examples: Parks, Dive spots, Birdwatching Areas, Zoos

B. NEGATIVE IMPACT OF RECREATION

- Unwanted effects on resources or experiences due to recreation
- **Soil and Water:**
 - Trampling leads to soil compaction, erosion, and increased water runoff.
- **Sanitation Problems:**
 - Improper drainage and unsuitable solid waste disposal.
- **Pathogens:**

- Contaminated water supply in swimming areas.

C. RECREATIONAL WATER ENVIRONMENTS

- Oceans, Lakes, Rivers, Pools, Spas, etc.

Hazard	Adverse Outcome
Currents/Tides	Drowning
Hard surfaces/ Animals	Impact Injury
Contaminated water	Disease outbreaks

Table 3. Hazards and Adverse Health Outcomes

D. MANAGEMENT OF RECREATIONAL AREAS

- Ensure proper drainage and sewage.
- Maintain kitchen and refrigeration hygiene.
- Control solid waste disposal.
- Educational approaches for visitors.

E. RECREATION IN THE AGE OF COVID-19

- Virus spreads via respiratory droplets.
- **Outside Benefits:**
 - Air movement disperses droplets; sunlight provides Vitamin D.
- **Guidelines:**
 - Social distancing, local mask mandates, and hand hygiene.

F. MANAGEMENT AND COVID-19 CONSIDERATIONS

- **Management Focus:**
 - Ensure proper drainage, sewage facilities, and food refrigeration.
- **SARS-CoV-2:**
 - Chlorine or bromine is sufficient to disinfect water.
- **Training:**
 - Importance of physical fitness and visitor safety education.

IV. CONCLUSION

- Our health is a dynamic state resulting from adaptations to our environment.

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